

Logistic regression interpretation

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Recap: logistic regression model

Y_i = dengue status (0 = negative, 1 = positive)

Age_i = age (in years)

Random component: $Y_i \sim \text{Bernoulli}(\pi_i)$

Systematic component: $\log\left(\frac{\pi_i}{1 - \pi_i}\right) = \beta_0 + \beta_1 Age_i$

Recap: interpretation

$$\log\left(\frac{\hat{\pi}_i}{1 - \hat{\pi}_i}\right) = -2.45 + 0.22 \text{ Age}_i$$

Slope:

- + A one-year increase in age is associated with an increase in the *log odds* of dengue of 0.22.
- + A one-year increase in age is associated with an increase in the *odds* of dengue by a *factor* of $e^{0.22} = 1.25$.

Intercept:

- + when age is 0, the estimated log odds of dengue is -2.45
- + when age is 0, the estimated odds of dengue is $e^{-2.45} = 0.086$

Recap: adding more variables

Now let's add WBC as a variable to the model:

```
m2 <- glm(Dengue ~ Age + WBC, data = dengue,  
          family = binomial)  
summary(m2)
```

$$\log\left(\frac{\hat{\pi}_i}{1 - \hat{\pi}_i}\right) = 0.34 + 0.15 \text{ Age}_i - 0.31 \text{ WBC}_i$$

A one-year increase in age is associated with an increase in the odds of dengue by a factor of $e^{0.15} = 1.16$, holding white blood cell count fixed.

Adding more variables

What about a *categorical* predictor?

```
m3 <- glm(Dengue ~ Age + Abdominal, data = dengue,  
          family = binomial)  
summary(m3)
```

$$\log\left(\frac{\hat{\pi}_i}{1 - \hat{\pi}_i}\right) = -0.25 + 0.22 \text{ Age}_i + 0.21 \text{ Abdominal}_i$$

How should I interpret the coefficient on abdominal pain?

Interpretation

$$\log\left(\frac{\hat{\pi}_i}{1 - \hat{\pi}_i}\right) = -2.45 + 0.22 \text{ Age}_i$$

Slope:

- + A one-year increase in age is associated with an increase in the *log odds* of dengue of 0.22.
- + A one-year increase in age is associated with an increase in the *odds* of dengue by a *factor* of $e^{0.22} = 1.25$.

Can I interpret the coefficient in terms of the *probability* (rather than odds or log odds)?